

Basketball Shot Category Recognition Using Wearable Sensors and Feature-Based Machine Learning

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Abstract—Basketball shooting performance is traditionally evaluated through expert coaching or vision-based systems, both of which are subjective, costly, and difficult to scale. Recent advances in wearable sensing enable a low-cost alternative for objective and automated skill assessment. This paper presents a wearable sensor-based machine learning framework for classifying three fundamental basketball shot categories distinguished by court location: short-range (paint), mid-range (free-throw line), and long-range (three-point zone) shots. Sensor data were collected from smartphones worn on the forearm of eleven participants with diverse anthropometric characteristics. A robust preprocessing pipeline was implemented, incorporating signal interpolation to ensure consistent sampling, rotation-invariant magnitude computation, and adaptive peak-based shot segmentation. Statistical and frequency-domain features were extracted from multi-sensor inertial data and used to train several machine learning classifiers, including k-Nearest Neighbors, Support Vector Machines, Random Forest, CatBoost, and Neural Networks. Experimental results demonstrate that ensemble-based machine learning models outperform deep learning approaches in this limited-data scenario, achieving high classification accuracy with low inference latency suitable for real-time feedback. The findings confirm the feasibility of non-intrusive, wearable-based basketball skill assessment systems and establish a practical foundation for future data-driven and personalized training tools.

I. INTRODUCTION

Sports and physical activity are essential for maintaining health and developing motor and cognitive skills. Among team sports, basketball is one of the most popular worldwide, with shooting accuracy playing a decisive role in competitive performance. Effective evaluation of shooting skills is therefore a key component of basketball training.

Shooting proficiency depends on biomechanical factors such as coordination, timing, release angle, and force generation. Traditionally, shooting assessment relies on expert coaches who visually observe players and provide qualitative feedback. While effective, this approach is subjective, time-consuming, and difficult to scale, particularly for amateur players or individuals training independently. Camera-based tracking systems offer quantitative analysis but are costly and impractical for everyday training environments.

From a physical standpoint, a basketball shot can be modeled as a projectile motion problem, where different shooting distances require distinct force and release characteristics. As a result, short-range (paint), mid-range (free-throw line), and long-range (three-point zone) shots exhibit measurable differences in movement dynamics. Automatically distinguishing between these shot categories is a crucial step toward objective evaluation of shooting mechanics.

Recent advances in wearable sensing and machine learning have enabled automated recognition of human activities using inertial sensors. However, existing basketball-related studies primarily focus on general activity recognition or generic shot detection, often relying on deep learning models that require large datasets. The classification of specific basketball shot categories using wrist- or forearm-mounted wearable sensors remains relatively unexplored, particularly in limited-data scenarios.

To address this gap, this paper presents a wearable sensor-based machine learning framework for classifying three fundamental basketball shot categories: paint, free-throw line, and three-point zone shots. Using inertial data collected from smartphones worn on the forearm, a robust preprocessing and feature-engineering pipeline is applied, followed by an evaluation of multiple machine learning classifiers. The results demonstrate that feature-based and ensemble machine learning models can accurately distinguish between shot categories with low inference latency, establishing a practical foundation for scalable and automated basketball skill assessment.

The main contributions of this work are summarized as follows:

- The creation of a multi-sensor basketball shooting dataset covering three distinct shot distances.
- The development of a feature-based machine learning pipeline for shot-type classification.
- An experimental analysis demonstrating the effectiveness of classical machine learning models on limited sensor data.

II. RELATED WORK

The use of wearable inertial sensors combined with machine learning has become increasingly common in sports analytics due to their low cost, portability, and ability to capture biomechanical motion in real-world environments [1], [2]. Accelerometers and gyroscopes form the basis of many human activity recognition (HAR) systems and have been successfully applied across a wide range of physical activities.

In basketball, wearable-sensor-based studies have focused mainly on recognizing general activities such as running, jumping, dribbling, and shooting [3], [4]. These works demonstrate that inertial data contain distinctive motion patterns related to basketball movements. However, shooting is typically treated as a single activity class, without differentiation based on shooting distance or biomechanical demand, limiting fine-grained skill assessment.

Classical machine learning approaches, including k-Nearest Neighbors (KNN), Support Vector Machines (SVM), and Random Forests, are widely used with handcrafted time- and frequency-domain features for inertial-based activity recognition [1], [5]. Ensemble methods such as Random Forests have shown strong robustness and generalization performance in small datasets, which are common in sports-related experiments [6]. In basketball-specific research, most classical machine learning studies focus on detecting shot execution rather than classifying shot types or distances [4], [7].

More recently, deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have been applied to wearable-sensor-based activity recognition [2], [8]. Although these methods can automatically learn discriminative features, they typically require large labeled datasets and substantial computational resources. In limited-data scenarios, deep learning models often suffer from overfitting, making classical feature-based approaches more reliable [3], [9].

From a biomechanical perspective, prior studies have shown that shooting distance significantly influences force generation, release angle, and movement coordination in basketball jump shots [10], [11]. Despite this, biomechanical differences across shot distances have not been sufficiently exploited in wearable-sensor-based classification systems.

In summary, while wearable sensors and machine learning have been successfully applied to basketball activity recognition, there is a lack of research on classifying specific basketball shot distances using lightweight, sensor-based models. This work addresses this gap by proposing a feature-based machine learning framework capable of distinguishing Paint, Free Throw Line, and 3-Point Zone shots using forearm-mounted inertial sensors.

III. DATA COLLECTION AND PROPOSED SYSTEM

A. System Architecture

The overall system pipeline consists of several stages, including basketball shooting execution, sensor data acquisition, preprocessing and feature extraction, machine learning classification, and shot-type prediction.

Figure 1 illustrates the block diagram of the proposed system, spanning data acquisition to final classification.

B. Data Collection Setup

Data were collected from eleven participants, including three females and eight males, with heights ranging from 159 cm to 196 cm. Each participant performed thirty shots: ten paint shots, ten free-throw line shots, and ten three-point zone shots, resulting in a balanced dataset across all classes.

A smartphone was securely strapped between the wrist and elbow of the shooting arm, oriented outward and facing away from the body. The data collection setup, including the forearm-mounted smartphone sensor used during shooting, is illustrated in Fig. 2.

This placement ensured consistent capture of arm motion during shooting. The mobile device recorded data from multiple sensors, including the accelerometer, gyroscope, gravity sensor, and orientation sensor.

Participants were instructed to briefly keep their arm still after each shot to facilitate separation of individual shot motions. Although minor variations occurred due to natural movement and strap adjustment, the majority of recordings contained clean and distinguishable shot patterns.

The spatial orientation and representative trajectories of the three shot categories—Paint, Free Throw, and 3-Point Zone—are illustrated in Fig. 3, demonstrating the distinct projectile motion dynamics the system is designed to classify.

C. Preprocessing, Segmentation, and Feature Extraction

Raw sensor data were synchronized and resampled to a fixed sampling rate of 50 Hz ($T_s = 20$ ms) to ensure temporal consistency across all sensor streams. To isolate biomechanical markers such as vertical lift energy, raw inertial data were transformed from the device-local coordinate system to a global Earth-fixed reference frame using quaternion-based rotation, aligning the vertical axis with the direction of gravity. Following this transformation, both axis-aligned acceleration components and a rotation-invariant representation were retained to ensure robustness to sensor tilting during dynamic movement.

The total acceleration magnitude was computed as:

$$A_{mag} = \sqrt{A_x^2 + A_y^2 + A_z^2}, \quad (1)$$

providing an orientation-independent measure of movement intensity.

Shot segmentation was performed using a threshold-based peak detection algorithm applied to the acceleration magnitude signal. When A_{mag} exceeded a predefined activity threshold (e.g., 11.5 m/s^2), the event was identified as a candidate shot and a dynamic window centered around the detected peak was extracted. Windows exhibiting insufficient variance were discarded as noise or pauses, ensuring that only valid shot executions were retained for analysis.

Each segmented window was represented using a feature-based approach, as raw sensor signals are unsuitable for

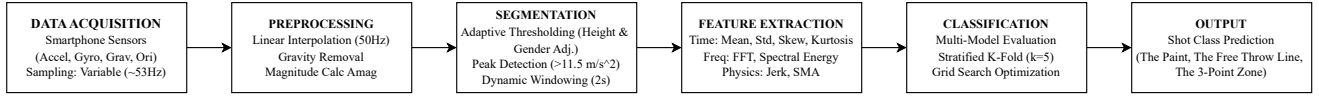


Fig. 1. Proposed Biomechanical Machine Learning Pipeline: From Multi-Modal Sensor Acquisition to Shot Classification.



Fig. 2. Data collection setup illustrating a participant wearing the forearm-mounted smartphone sensor during a basketball shot.

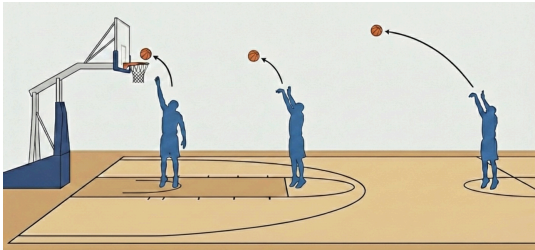


Fig. 3. Illustration of basketball shot trajectories for the three study categories: short-range (Paint), mid-range (Free Throw), and long-range (3-Point Zone). The diagram highlights the variations in release angles and force requirements that generate the distinct biomechanical markers captured by the forearm-mounted sensors.

direct classification with classical machine learning models. Time-domain statistical features, including mean, standard deviation, minimum, maximum, skewness, and kurtosis, were extracted for each sensor axis. Frequency-domain features were computed using the Fast Fourier Transform (FFT) to

capture dominant frequency components and spectral energy.

In addition to standard signal statistics, domain-specific biomechanical features were engineered to better characterize shooting mechanics. These included jerk to represent motion smoothness, an effort ratio defined as acceleration intensity relative to height-based biomechanical expectations, and an estimated release angle derived from the device pitch orientation at the moment of peak acceleration. Furthermore, magnitude-based energy features were calculated by integrating the acceleration magnitude over the window to approximate the force exerted during shot execution.

All extracted features were concatenated into a single feature vector representing one shot instance. Participant anthropometric attributes, namely height and gender, were appended to the feature vector to account for biomechanical variability across subjects.

IV. EXPERIMENTAL RESULTS

We tested the proposed framework on a dataset of 330 basketball shots collected from 11 participants, where each participant performed 30 shots (10 paint shots, 10 free-throw line shots, and 10 three-point zone shots). After preprocessing the inertial signals using interpolation for consistent sampling, magnitude computation to reduce rotation effects, and adaptive peak detection for accurate shot segmentation, a set of statistical and frequency-domain features was extracted. These features included mean, variance, skewness, spectral entropy, and dominant frequency components.

The dataset was divided into 80% training and 20% testing sets. Grid-search hyperparameter tuning was performed for each model to identify the best configuration. A total of nine classifiers were evaluated: k-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, Random Forest, LightGBM, XGBoost, CatBoost, Support Vector Classifier (SVC), and a Multi-Layer Perceptron (MLP) neural network. Performance was primarily assessed using classification accuracy and inference latency (measured in milliseconds per sample), as real-time feedback during training requires fast predictions.

The final leaderboard, ranked by test accuracy, is presented in Table I.

The Support Vector Classifier (SVC) achieved the best overall performance, reaching 95.45% test accuracy with negligible inference latency, making it the most effective model in terms of both precision and computational efficiency. The optimized MLP followed closely, achieving 93.94% accuracy with sub-millisecond inference time. Ensemble-based methods such as CatBoost and Random Forest also performed strongly; however, Random Forest exhibited significantly higher inference latency, making it less suitable for real-time deployment.

TABLE I
PERFORMANCE COMPARISON OF EVALUATED MODELS

Model	Accuracy (%)	Latency (ms/sample)
SVC	95.45	0.9383
Neural Network (MLP)	93.94	0.9380
CatBoost	92.42	1.0305
k-Nearest Neighbors (KNN)	90.91	1.8361
Random Forest	90.91	18.1621
LightGBM	87.88	1.1271
XGBoost	84.85	2.4243
Logistic Regression	83.33	0.9576
Decision Tree	68.18	1.3349

Simpler models, such as the Decision Tree, showed noticeably lower accuracy, likely due to overfitting on the relatively small dataset.

A scatter plot of accuracy versus latency further highlights the trade-off between prediction performance and computational cost. The SVC and MLP models occupy the top-left region of the plot, indicating high accuracy and low latency, whereas Random Forest achieves competitive accuracy at the expense of substantially slower inference. Confusion between shot categories was minimal among the top-performing models, detailed class-wise classification results for the best-performing SVC model are provided in Appendix A., confirming that the extracted features effectively captured biomechanical differences in force generation, timing, and arm motion across short-, mid-, and long-range basketball shots.

V. CONCLUSION

This work demonstrates that a simple, forearm-worn smartphone sensor combined with classical machine learning techniques can reliably distinguish between three fundamental basketball shot categories: paint (short-range), free-throw line (mid-range), and three-point zone (long-range), achieving very high classification accuracy.

Among the evaluated models, the Support Vector Classifier (SVC) emerged as the most effective approach, achieving 95.45% test accuracy with negligible inference latency. Ensemble-based methods, such as CatBoost and Random Forest, as well as a lightweight neural network, also delivered strong performance. However, deep learning approaches were not required in this setting; instead, feature-based classical machine learning models proved more effective and computationally efficient given the limited dataset size.

These results highlight a practical and low-cost solution for objective, automated basketball skill assessment that does not rely on expensive camera systems or subjective coach evaluations. The proposed framework enables real-time feedback and has the potential to support training tools that help players—particularly amateur athletes training independently—better understand and improve their shooting mechanics based on distance-specific movement patterns.

Future work will focus on expanding the dataset to include a larger and more diverse population, as well as shots performed under fatigue, pressure, or defensive conditions. Further extensions may involve integrating the proposed model into

a mobile application for on-court feedback or incorporating additional wearable sensors. Overall, this study confirms that wearable inertial sensing combined with effective feature engineering offers a scalable and accessible pathway for bringing data-driven insights into everyday basketball training.

APPENDIX A
CONFUSION MATRIX OF THE SVC MODEL

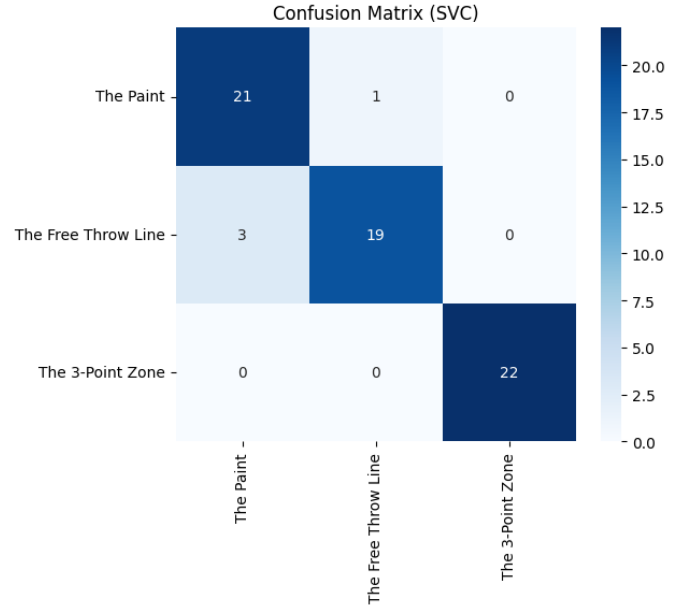


Fig. 4. Confusion matrix of the Support Vector Classifier (SVC) on the test dataset.

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